



CATHOLIC JUNIOR COLLEGE
JC2 Preliminary Examinations
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

--

CLASS

2T		
----	--	--

INDEX
NUMBER

--	--	--	--

BIOLOGY

9648/01

Paper 1 Multiple Choice

1 September 2015

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your index number and name on the Answer Sheet in the spaces provided.

There are forty questions on this paper. Answer all questions. For each question there are four possible answers A, B, C and D.

Choose the one you consider correct and record your choice in soft pencil in the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

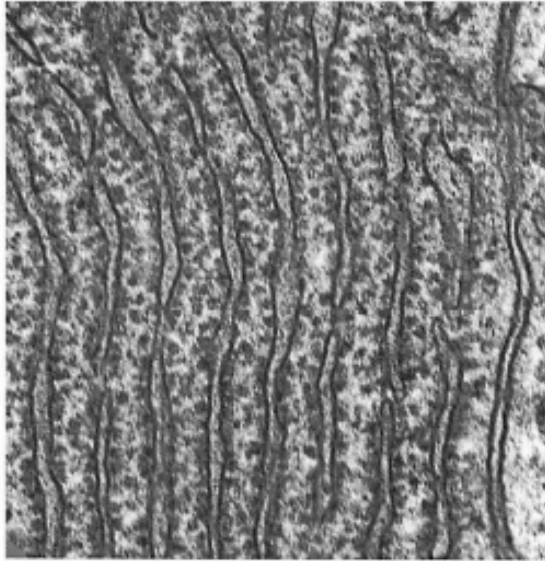
Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

This document consists of **26** printed pages.

[Turn over

- 1 The electronmicrograph shows part of an organelle in a cell.



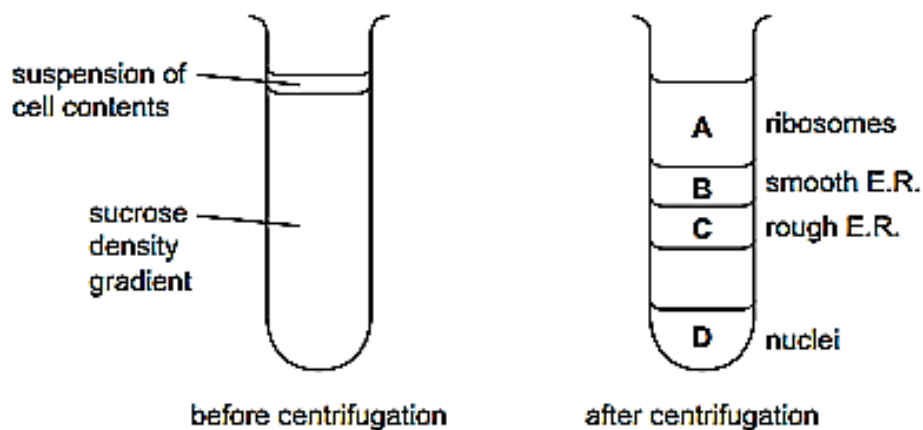
What describes a function of the tubules in the organelle shown?

- A** moving proteins to places where they are covered by phospholipid membranes for secretion outside the cell.
- B** modifying proteins and covering them with phospholipid membranes for secretion outside the cell
- C** producing proteins, covering them with phospholipid membranes and moving them for use inside the cell
- D** producing ribosomes and proteins and storing them in phospholipid membranes for use inside the cell

- 2 Sometimes scientists need to isolate organelles. This can be achieved by taking a number of cells and breaking their cell surface membranes to release the contents of the cells into a buffer solution.

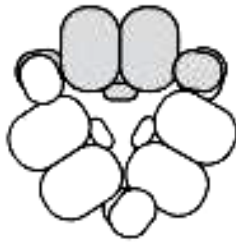
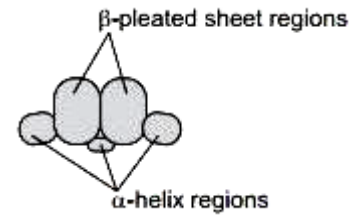
In zonal centrifugation, the suspension of cell contents is placed on top of a sucrose density gradient. The tube is then placed in a centrifuge and spun at high speed. The heavier particles will move towards the bottom of the tube faster than lighter particles, as shown below.

If a sample of intact prokaryotes had been added to a suspension of eukaryotic cell contents, where would you expect them to be found?



3 Approximately half of the total protein in a pea seed consists of the storage protein vicilin.

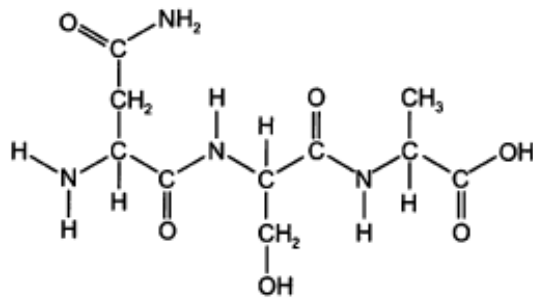
- Each molecule of vicilin is made up of three identical polypeptides.
- Each polypeptide is made up of two β -pleated sheet regions with linking α -helix regions, folded into the shape shown to the right.
- This allows the three polypeptides to pack together into a compact, flat storage molecule, as shown below.



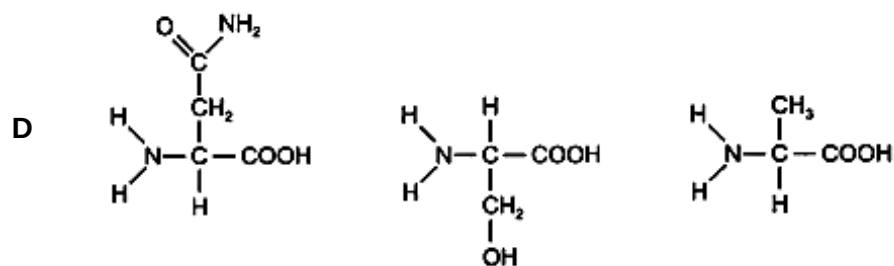
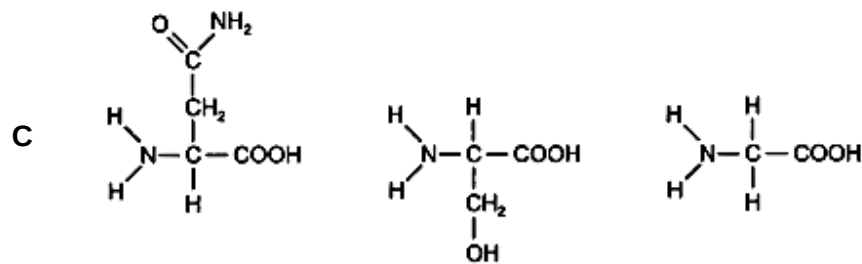
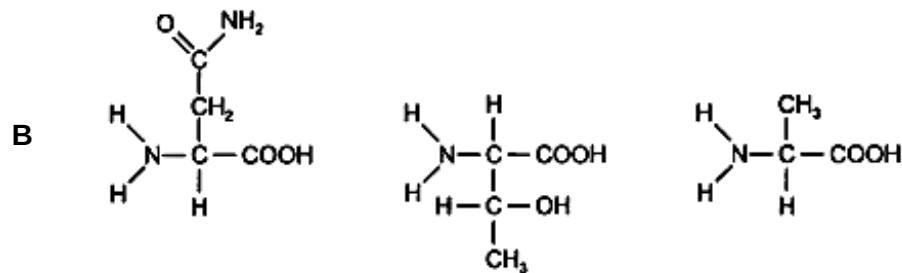
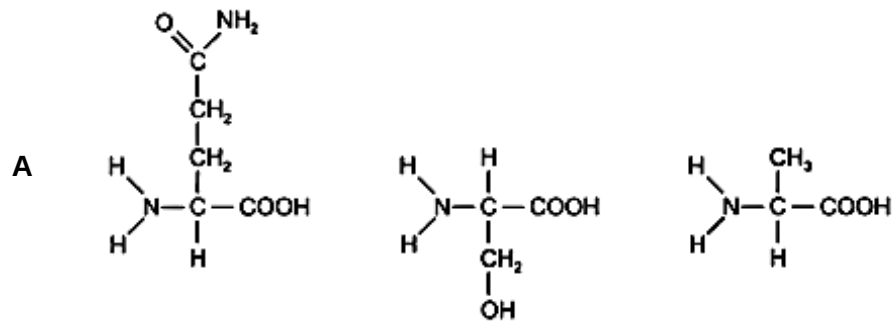
Which row correctly describes the structure of vicilin?

	primary structure	secondary structure	tertiary structure	quaternary structure
A	amino acid sequence of one polypeptide	α -helix and β -pleated sheet regions of each polypeptide	association of three polypeptides	folding of each polypeptide
B	amino acid sequence of one polypeptide	α -helix and β -pleated sheet regions of each polypeptide	folding of each polypeptide	association of three polypeptides
C	association of three polypeptides	amino acid sequence of one polypeptide	α -helix and β -pleated sheet regions of each polypeptide	folding of each polypeptide
D	association of three polypeptides	amino acid sequence of one polypeptide	folding of each polypeptide	α -helix and β -pleated sheet regions of each polypeptide

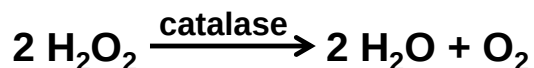
- 4 The diagram shows the molecular structure of a peptide.



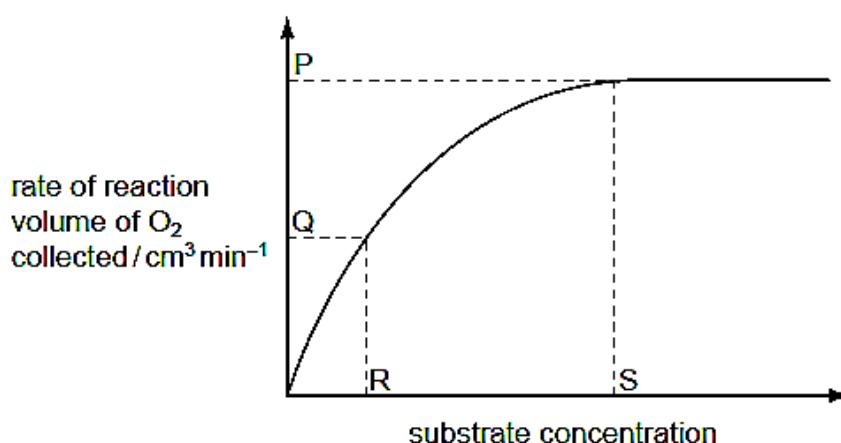
Which molecules would result from the complete hydrolysis of the peptide?



- 5 The diagram shows the action of an enzyme called catalase, produced by liver tissue, which breaks down hydrogen peroxide into water and oxygen.



The rate of this reaction can be determined by measuring the volume of oxygen produced in a given length of time. Students added small cubes of fresh liver tissue to a range of hydrogen peroxide solutions and measured the volumes of oxygen produced. Their data were used to produce the graph showing how changing the concentration of hydrogen peroxide affected the rate of oxygen production.



Which statements are correct?

- I At P, the rate of reaction is limited by the concentration of enzyme.
 - II At Q, all of the enzyme active sites are occupied by substrate molecules.
 - III At Q, the rate of reaction is limited by the concentration of the substrate.
 - IV At S, all of the enzyme active sites are occupied by substrate molecules.
- A I and IV
B II and III
C II and IV
D I, III and IV

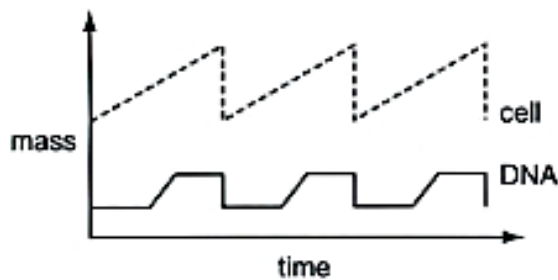
- 6 The amount of DNA in a mammalian cell in early prophase I of meiosis is $4\mu\text{g}$. What is the amount of DNA in the same cell at G1 of interphase?

A $1\mu\text{g}$
 B $2\mu\text{g}$
 C $4\mu\text{g}$
 D $8\mu\text{g}$

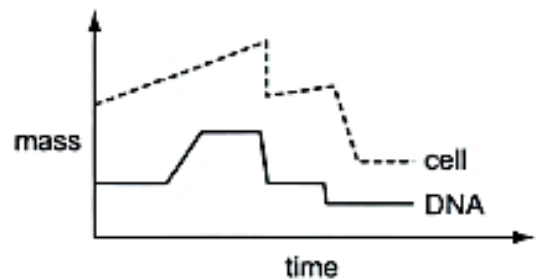
- 7 Cell division is the means of almost all growth and reproduction.

Which graph correctly represents a form of cell division that maintains genetic stability at the expense of variation?

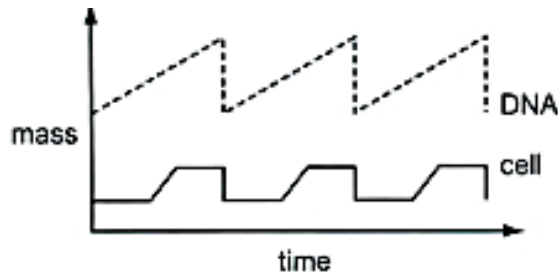
A



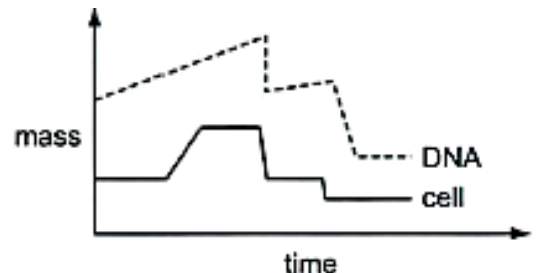
B



C



D



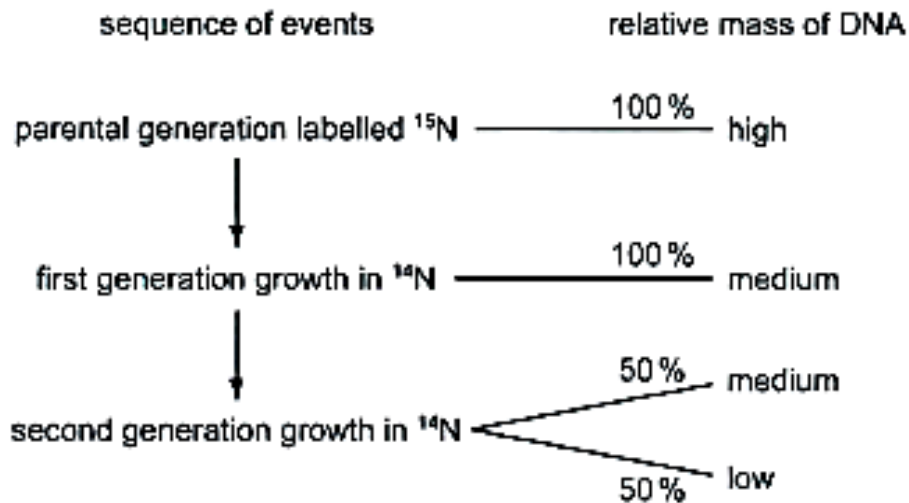
- 8 Which of the following statements about replication is **false**?

A No ATP is utilized in the formation of phosphodiester bonds on the leading strand.
 B DNA polymerase I and III both need a free $3'\text{OH}$ to carry out their function.
 C Synthesis of the lagging strand occurs in a $3'$ to $5'$ direction.
 D On the lagging strand, primase is blocked by the action of helicase thus forming Okazaki fragments.

- 9 Messelson and Stahl investigated DNA replication by growing bacteria in culture containing heavy nitrogen, ^{15}N , until all the DNA was labelled.

These bacteria, the parental generation, were then transferred to a medium containing only light nitrogen, ^{14}N and allowed to replicate for two generations.

DNA was extracted from each generation of bacteria and its relative mass estimated. The flow diagram shows the results.

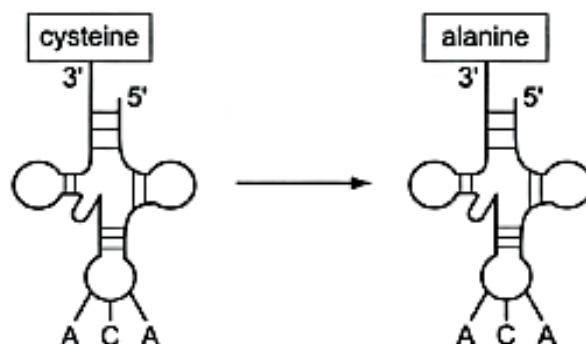


Which row explains the mass of DNA in the second generation grown in ^{14}N ?

	50% DNA molecules		50% DNA molecules	
	strand 1	strand 2	strand 1	strand 2
A	only ^{15}N	only ^{14}N	only ^{14}N	only ^{14}N
B	only ^{15}N	only ^{15}N	only ^{14}N	only ^{14}N
C	only ^{15}N	only ^{15}N	only ^{14}N	only ^{15}N
D	only ^{15}N	half ^{14}N , half ^{15}N	only ^{14}N	half ^{14}N , half ^{15}N

- 10** Transfer RNA combined with an amino acid is called amino-acyl tRNA. It is possible to chemically convert the amino acid cysteine into the amino acid alanine whilst it is still attached to its tRNA.

The altered amino-acyl tRNA still binds to UGU triplets on messenger RNA (mRNA), but now incorporates alanine into the resulting polypeptide instead of cysteine.



Which statement is correct?

- A** A codon on the amino-acyl tRNA determines its specificity.
- B** Both the amino acid and the anticodon of an amino-acyl tRNA affect where it binds to mRNA.
- C** The amino acid of an amino-acyl tRNA does not influence its binding to mRNA.
- D** The codon-anticodon interaction is influenced by the amino acid on an amino-acyl tRNA.
- 11** The petals of the morning glory flower are purple when the flower is in the bud but become blue as the flower opens. This colour change is caused by an increase in pH of the vacuoles of the petal cells.

A mutation in a gene coding for a type of hydrogen ion (H^+) pump in the vacuole membranes of the petal cells results in flowers that are purple when open.

The difference in DNA sequence of the gene coding for the ion pump in plants with blue or purple flowers is shown below.

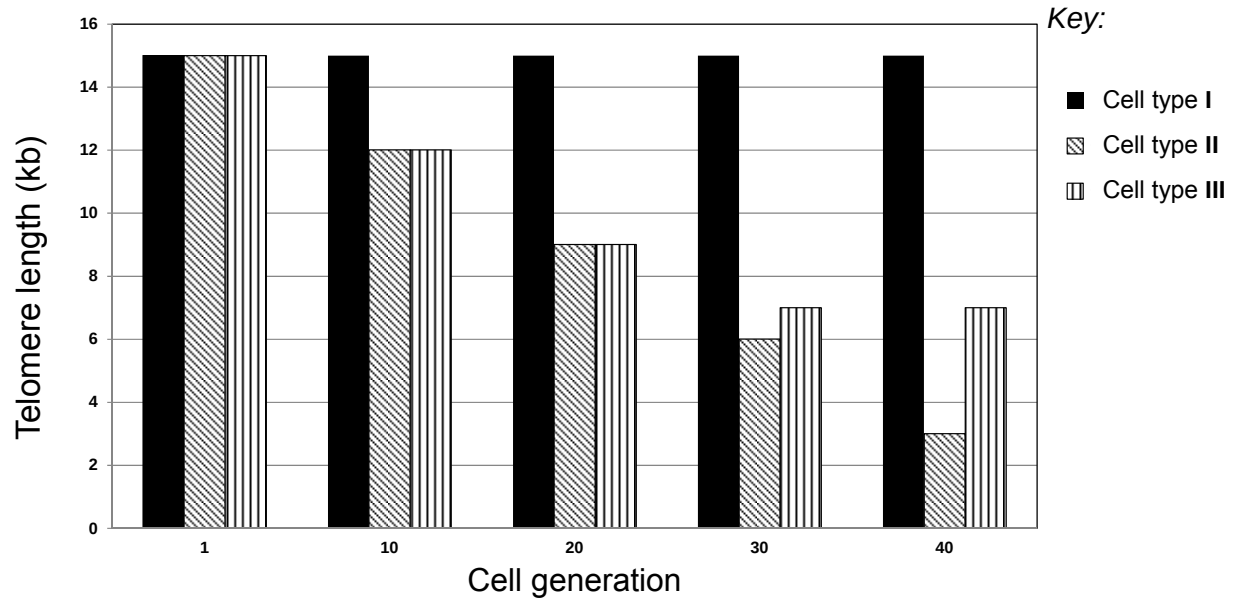
Plant with blue flowers: -T-T-A-A-T-C-C-T-G-A-G-A-T-T-T-

Plant with purple flowers: -T-T-A-A-T-C-C-T-G-C-T-G-A-G-A-T-T-T

Which statements explain the purple colour of the flowers of the mutant plant?

- 1 A frameshift mutation alters the sequence of amino acids in the ion pump.
 - 2 The primary structure and the shape of the ion pump are altered.
 - 3 The activity of the ion pump is altered by changes in its shape.
 - 4 The mutant ion pump cannot pump H^+ ions into the vacuoles of the petal cells.
- A** 1 and 2
- B** 1 and 4
- C** 2 and 3
- D** 3 and 4

12 The graph shows the telomere length of three different cell types I to III.



Which of the following correctly identifies cell types I to III at generation 40?

	Cell type		
	I	II	III
A	Stem cells	Somatic cells	Tumour cells
B	Stem cells	Tumour cells	Somatic cells
C	Germline cells	Somatic cells	Stem cells
D	Somatic cells	Tumour cells	Germline cells

- 13** The diagram represents a length of DNA from a prokaryote that includes a structural gene. Parts of the length of DNA are labelled **W**, **X** and **Y**. They have different functions in the control of transcription of the structural gene.



What identifies the functions of parts **W**, **X** and **Y**?

	W	X	Y
A	operator	regulator	regulator
B	promoter	regulator	operator
C	regulator	promoter	operator
D	promoter	operator	promoter

- 14** A single mutation has occurred in an *E. coli* cell. If the structural genes in a *trp* operon were not expressed regardless of the presence or absence of the amino acid tryptophan, which of the following mutations could have taken place?

- 1 The cell had a mutation at the operator of the *trp* operon.
- 2 The cell had a mutation at the *trpR* gene.
- 3 The cell had a mutation at the promoter of the *trp* structural genes

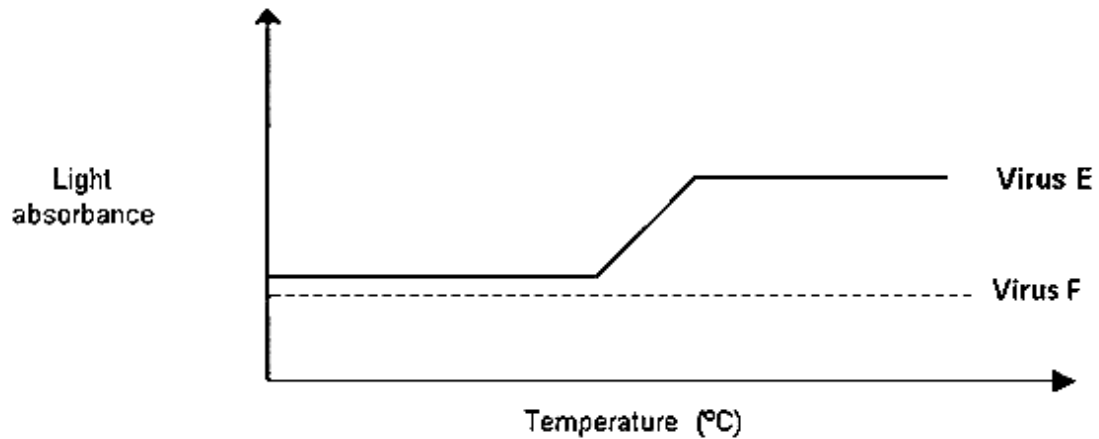
- A** 3 only
B 1 and 2
C 2 and 3
D 1, 2 and 3

- 15** Which of the statements are false about RNA viruses such as influenza and HIV?

- 1 The virus must encode either a RNA or a DNA polymerase.
- 2 The virus must contain a RNA polymerase.
- 3 The virus must encode receptor protein for entry into host cells.
- 4 The virus must encode proteases which can generate functional proteins from a polyprotein.

- A** 1 and 3
B 1 and 2
C 2 and 4
D 2, 3 and 4

- 16 Two new viruses **E** and **F**, which infect eukaryotic cells, have been identified. In one experiment, the nucleic acid from each virus is isolated and analyzed over a range of temperatures. The light absorbance of nucleic acids changes when denaturation or annealing occurs. The following graph shows the effect of temperature on the light absorbance of the nucleic acids found in viruses **E** and **F**.

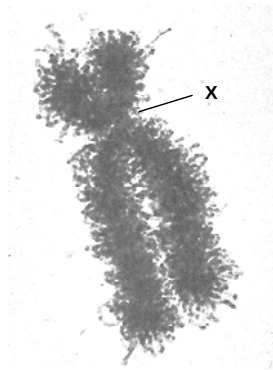


In a second experiment, it is found that treatment with RNA-dependent RNA polymerase inhibitors or with inhibitors of DNA synthesis blocks the ability of virus **F** to infect cells. In contrast, RNA-dependent RNA polymerase inhibitors have no effect on the ability of virus **E** to infect cells but DNA synthesis inhibitors block infection by virus **E**.

Which of the following conclusions can be drawn from both the experiments?

- A** The genome of virus **E** is single-stranded RNA and that of virus **F** is double-stranded DNA.
- B** The genome of virus **E** is double-stranded DNA and that of virus **F** is single-stranded RNA.
- C** The genome of virus **E** is double-stranded RNA and that of virus **F** is single-stranded DNA.
- D** The genome of virus **E** is double-stranded DNA and that of virus **F** is double-stranded RNA.

- 17 The electron micrograph shows a specialised region X found on a chromosome.



Which of the following statements are true of X?

- I With a mutation at X, there is no alignment of chromosomes at metaphase.
- II X is replicated during S phase.
- III X is transcriptionally inactive throughout the cell cycle.
- IV The length of X is highly conserved.

- A I and IV only
- B III and IV only
- C I, II and III only
- D All of the above

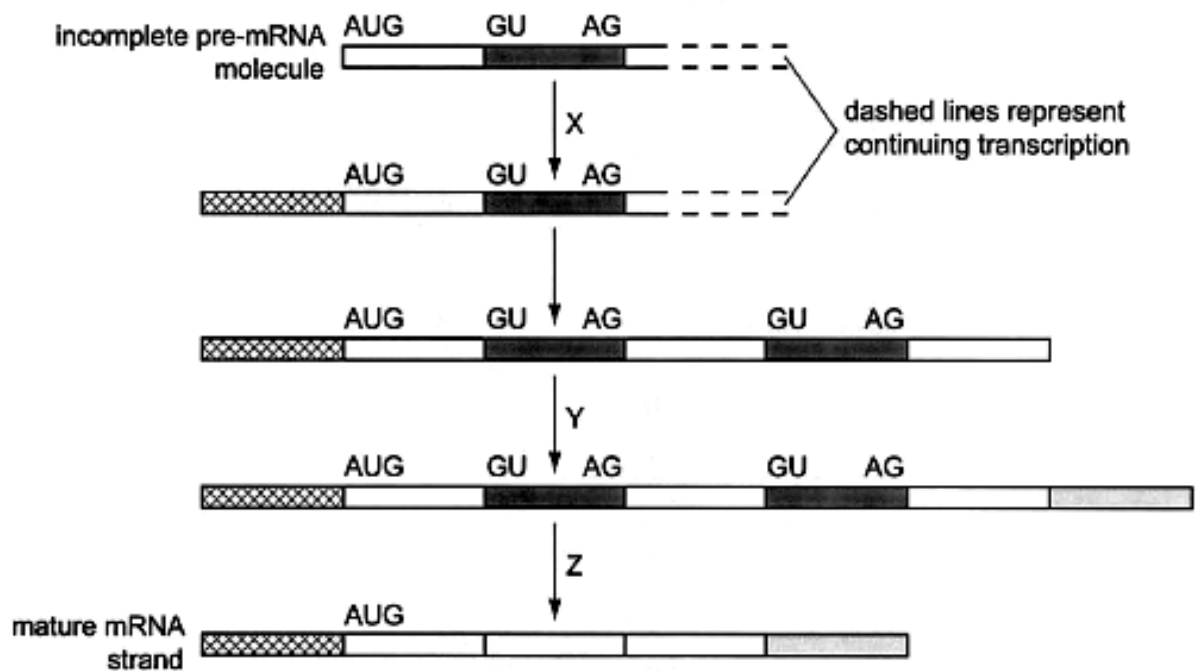
- 18 DNase is added to samples taken from the same chromosomal region from three different cell types. After allowing for digestion of DNA, the remaining DNA samples are weighed with the following results.

Cell type	Mass of DNA before adding DNase/ μg	Mass of DNA after adding DNase/ μg
A	5.0	5.0
B	5.0	3.4
C	5.0	4.1

What do these results suggest about genes and their expressions in all the three different cell types?

- A Some regions of the cell type A chromosomes contains RNA
- B The genes found in the same chromosomal region in all the three cell types are expressed the most in cell type B.
- C The mass of DNA in cell type A did not decrease because DNA is not of complementary shape to the active site of DNase.
- D There are more phosphodiester bonds in the DNA found in the chromosomes of cell type B than those in cell type A and C.

- 19 The diagram represents stages in transcribing and processing a pre-mRNA molecule to form a strand of mature mRNA



Which row correctly identifies processes X, Y and Z on the diagram?

	X	Y	Z
A	cap added	splicing	poly A added
B	cap added	poly A added	splicing
C	poly A added	cap added	splicing
D	splicing	cap added	poly A added

- 20 Which comparative statements about prokaryotic and eukaryotic gene expression are correct?

- 1 DNA methylation is a feature of prokaryotes but not eukaryotes.
- 2 Eukaryotes and prokaryotes both use ribosomes to translate mRNA.
- 3 Eukaryotes have introns, most prokaryotes do not.
- 4 Prokaryotes have genes organised into operons, most eukaryotes do not.

- A 1, 2 and 3
 B 1, 2 and 4
 C 1, 3 and 4
 D 2, 3 and 4

- 21** It has been suggested that breast cancer cells produce high levels of hydrogen peroxide. This causes connective tissue cells near the cancer cells to digest some of their mitochondria, releasing nutrients which feed the cancer cells.

Which observations made on breast cancer cells and connective tissue cells growing in tissue culture support this view?

- 1 Breast cancer cells grown alone produce hydrogen peroxide.
- 2 Treating breast cancer cells with hydrogen peroxide causes apoptosis.
- 3 Connective tissue cells grown with breast cancer cells have reduced mitochondrial activity.
- 4 Treating breast cancer cells with peroxidase increases cancer cell death.

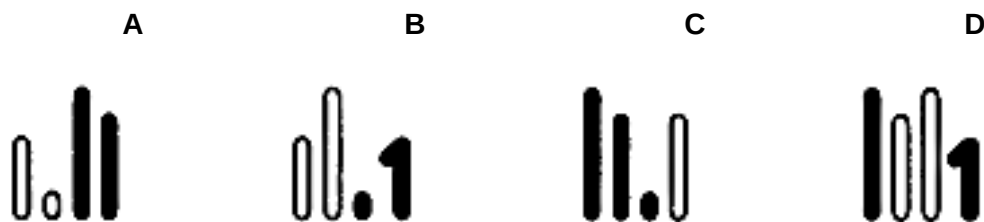
- A** 1, 2, 3 and 4
B 1, 3 and 4 only
C 1 and 4 only
D 2 and 3 only

- 22** No crossing over occurs during meiosis in male fruit flies of the species *Drosophila melanogaster*.

The diagram shows the four pairs of homologous chromosomes present in a testis cell of a male fly.



Which set of chromosomes in a gamete nucleus shows the genetic variation resulting from independent assortment?



23 Which statements about gene mutation are correct?

- 1 It can be brought about by exposure to ionising radiation.
- 2 It always produces dominant alleles.
- 3 It can increase the number of base pairs in a gene.
- 4 It can occur in both somatic and sex cells.

- A** 1 and 2 only
B 1 and 4 only
C 2 and 3 only
D 1, 3 and 4 only

24 Three gene loci in mice and the characteristics they control in mice are shown below.

Locus 1 (A/a) Determines coat colour	Locus 2 (K/k) Determines tail shape	Locus 3 (F/f) Determines condition of coat
agouti vs albino	kinky vs straight	frizzy vs non-frizzy

A student studying the inheritance of these characteristics carried out controlled crosses involving two loci at any one time. The outcomes of his experiments are shown below.

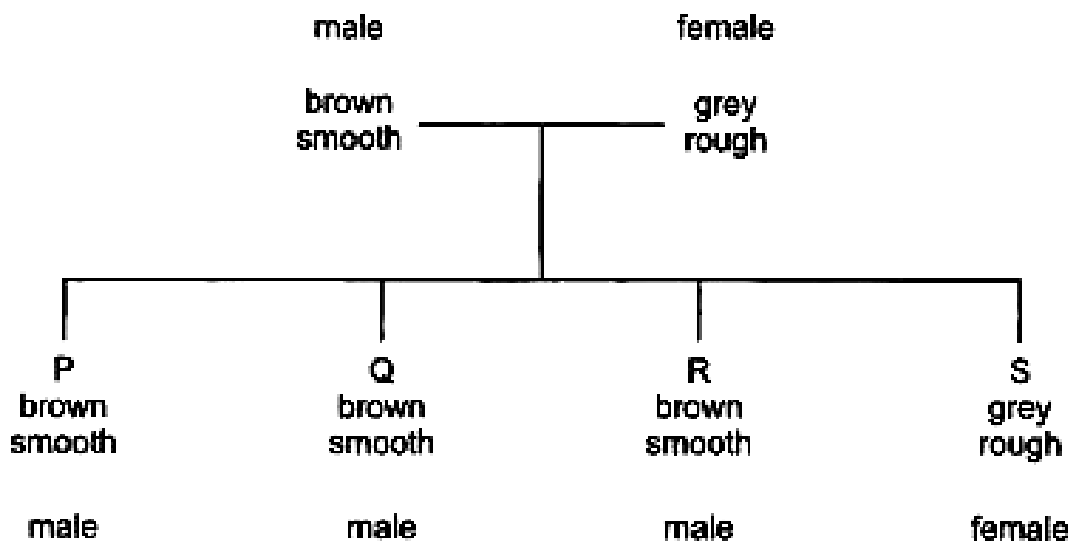
Parents (pure bred)	F1	Offspring of a test cross with F1
Cross 1 agouti, non-frizzy coat X albino, frizzy coat	agouti, non-frizzy coat	agouti, non-frizzy 44 albino, frizzy 46 agouti, frizzy 5 albino, non-frizzy 5
Cross 2 agouti, straight tail X albino, kinky tail	agouti, kinky tail	agouti, straight 23 albino, kinky 27 agouti, kinky 24 albino, straight 26

Which one of the following is a valid conclusion from his results?

- A** The genotype of the F1 of cross 1 is (Af)(aF).
B The map distance between albino and non-frizzy is 5 map units.
C The gene for coat colour and tail shape assort independently.
D Since there are more recombinants formed in cross 2, one possible arrangement of the 3 genes is agouti, non-frizzy, kinky.

- 25 In a species of mammal, one pair of alleles is responsible for the inheritance of hair colour and another pair controls hair texture. The allele for brown hair is dominant to the allele for grey hair and the allele for smooth hair is dominant to the allele for rough hair.

The diagram shows the results of crossing a male showing the dominant phenotype with a female showing the recessive phenotype.



If P is crossed with a female with the same genotype as P, what are the chances of the offspring having brown smooth hair?

- A 1 in 1 (100%) B 3 in 4 C 9 in 16 D 1 in 2

- 26 Which statement concerning chrysanthemum plants, of the genus *Dendranthema*, is a valid example of how the environment may affect the phenotype?

- A Anthocyanins and anthoxanthins are vacuolar pigments, whereas xanthophylls and carotenes are pigments found in membrane-bound organelles known as plastids. These, together with molecules known as co-pigments, are responsible for the variation observed in petal colour in *Dendranthema*.
- B Identical genetic crosses performed between varieties of *Dendeanthema* result in a greater proportion of offspring plants with plastids exhibiting a yellow colour when grown in a field and a greater proportion of offspring plants with colourless plastids when grown in a glasshouse.
- C The seeds of a cross between *Dendranthema weyrichii* and *Dendranthema grandiflora* produce plants that are far more frost-tolerant and exhibit an extended flowering season compared with both parent plants.
- D The seeds of a cross between *Dendranthema weyrichii* (height varying between 12.5 – 15.0 cm) and *Dendranthema grandiflora* (height varying between 8.0 – 25.0 cm) produce plants, when grown in natural day length, of a height varying between 55.0 – 71.0 cm.

- 27 A scientist is investigating the effects of Poison T on the cell signaling pathway of glucagon. It is found that Poison T diminishes the effect of glucagon.

The table below shows the different components of the cell signaling pathway of glucagon and their statuses.

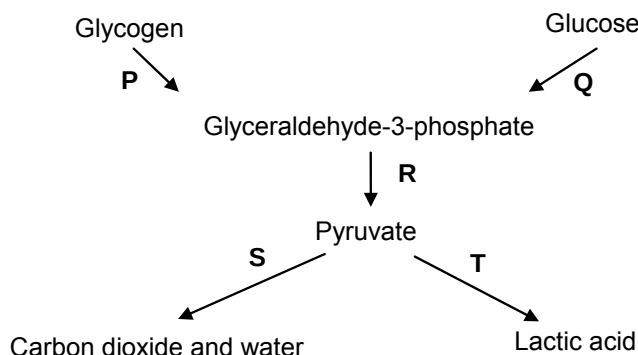
Component	Status
G-Protein Coupled Receptor (GPCR)	Configuration can be changed
G-protein	Can exchange GDP for GTP
cAMP levels	Low
Protein Kinase A	Inactivated

Which of the following statements can be concluded from the results?

- 1 Poison T interferes with reception.
- 2 Poison T prevents G-protein from hydrolysing GTP.
- 3 Poison T inactivates the enzyme adenylate cyclase.
- 4 Poison T stops the relaying of message down the phosphorylation cascade.

- A 1 and 2
B 2 and 3
C 2 and 4
D 3 and 4

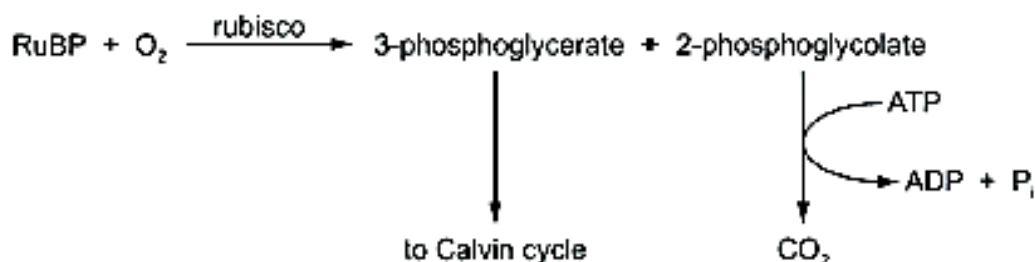
- 28 With reference to the diagram below, relate processes P, Q, R, S, T to statements (i), (ii) and (iii).



- (i) NAD is regenerated without the use of the electron transport system
(ii) ATP is synthesised via substrate level phosphorylation
(iii) It can take place under anaerobic conditions.

	(i)	(ii)	(iii)
A	R	S	T
B	T	R	Q
C	None of the above	S	R
D	S	Q	None of the above

- 29** Rubisco is the carbon dioxide-fixing enzyme. One rubisco molecule has eight active sites where carbon dioxide fixation occurs, with each active site catalysing only three reduction reactions per second.

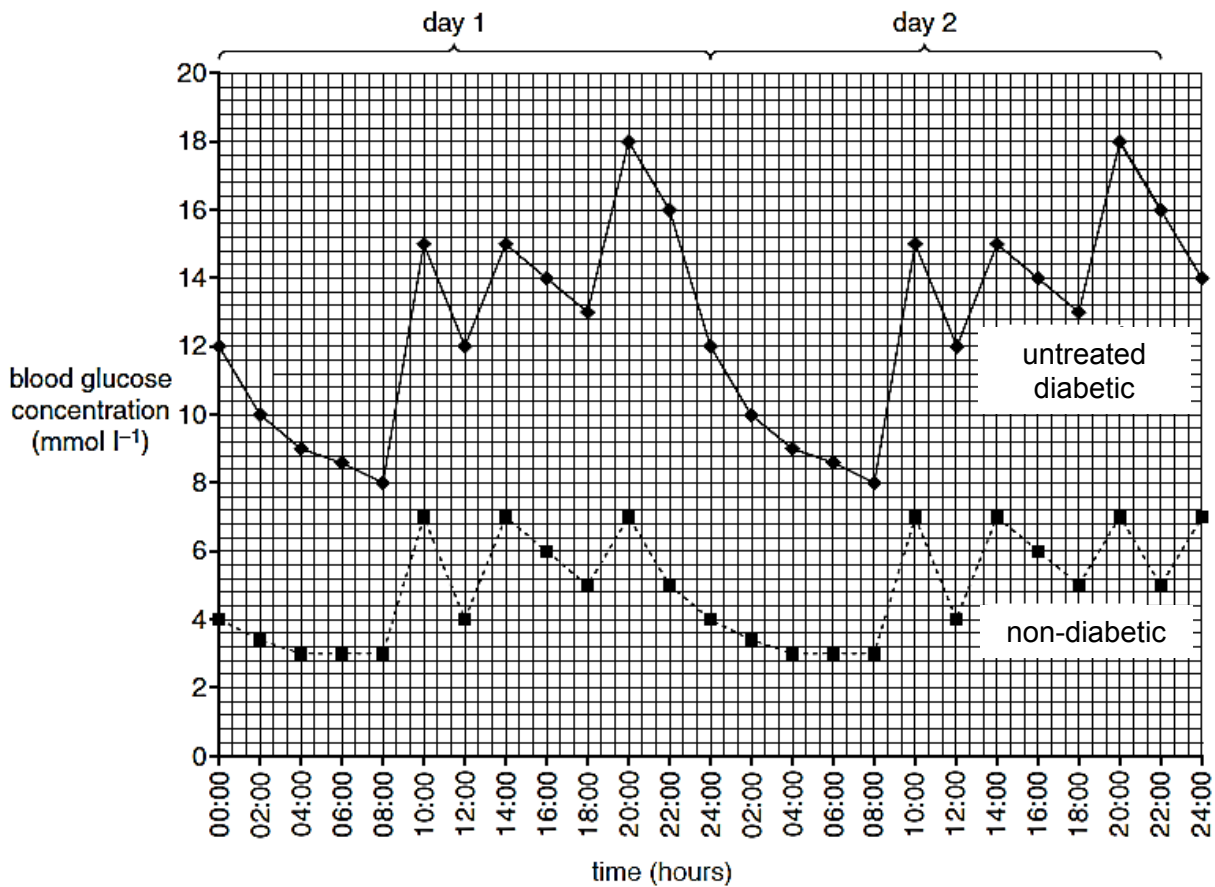


The enzyme also catalyses, at the same active sites, the addition of oxygen to ribulose biphosphate (RuBP). This reaction is favoured when oxygen concentrations in the leaf are high and carbon dioxide concentrations are low.

Which of the facts is paired with the correct explanation?

	fact	explanation
A	On very hot, dry days stomata close to prevent water loss.	This reduces the availability of oxygen, increasing the production of 3-phosphoglycerate.
B	Plants synthesise large volumes of rubisco.	This may be an adaptive response to compensate for low concentrations of oxygen.
C	Processing 2-phosphoglycolate will eventually release carbon dioxide.	This will increase the rate of reduction and increase the rate of RuBP regeneration, increasing the rate of photosynthesis.
D	Rubisco is an inefficient photosynthetic enzyme.	This is because the rate of carbon dioxide reduction can be decreased by the competitive binding of oxygen molecules to the active site.

- 30 The diagram shows the changes in blood glucose concentration over two days in an untreated diabetic and in a non-diabetic.



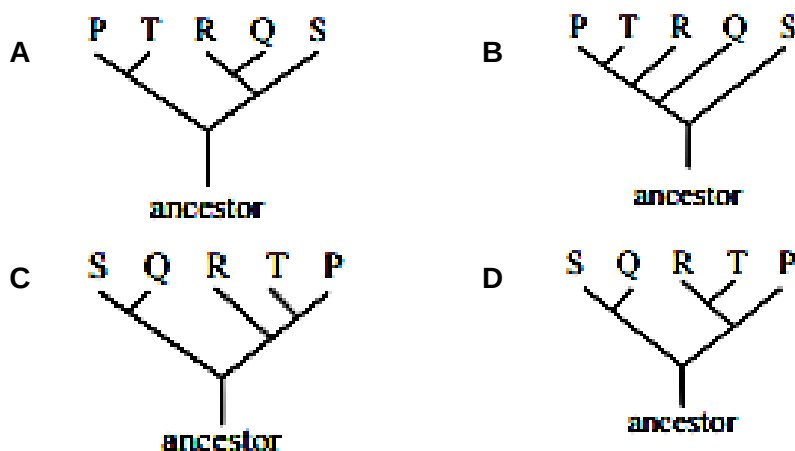
Which statements describing the diabetic could explain the differences between his blood glucose concentration and that of the non-diabetic?

- 1 Tissues are insensitive to insulin.
 - 2 Less insulin is produced by β cells of the islets of Langerhans in the pancreas.
 - 3 Less glucose uptake by cells occurs.
- A 1 and 2
 B 1 and 3
 C 2 and 3
 D 1, 2 and 3

- 31** In constructing an evolutionary tree, it is assumed that the more similarities shared between two species, the closer is their phylogenetic relationship. The amino acid sequences of a certain protein in five species of organisms were analysed and compared. The following table shows the number of amino acid differences between the amino acid sequence of species **P** and that of each of the other four species.

Species	Q	R	S	T
The number of amino acid differences between the amino acid sequence of species P and that of the other species	31	16	36	1

Assuming that these five species of organisms arose from the same ancestral stock, which of the following diagrams best represents the evolutionary tree constructed using the above information?



- 32** The pigment haemoglobin found in red blood cells of mammals and birds combines readily with oxygen.

DNA analysis has revealed that a form of haemoglobin is found in a wide range of unrelated phylogenetic groups including bacteria, annelids, arthropods and leguminous plants.

Which evolutionary processes could account for the distribution of haemoglobin in such a wide variety of organisms?

	adaptive radiation	conservation of genes	natural selection
A	✓	✓	✓
B	✓	✓	x
C	✓	x	✓
D	x	✓	✓

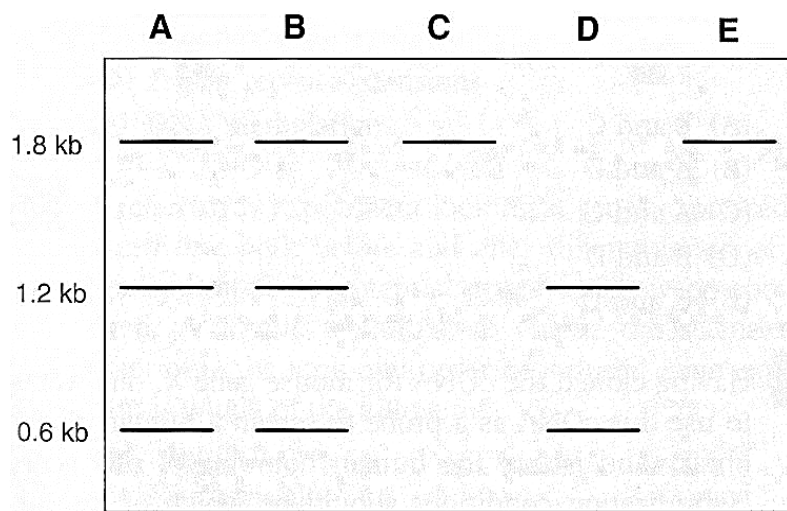
key

✓ = true

x = false

- 33 Which of the following is not a limitation of the use of fossil records as evidence for evolution?
- A Fossils are damaged and incomplete.
 - B Some organisms may not form fossils.
 - C Fossils are found in different sedimentary rock layers.
 - D Fossils present in inaccessible areas are not available to us for study.
- 34 A family exhibits a rare autosomal recessive disease, which is caused by a point mutation that abolishes an *Eco*R1 restriction site in the genome. To ascertain the genotype of family members, researchers did the following:
- (i) amplify a 1.8 kb region of this area using PCR;
 - (ii) cut the PCR products with *Eco*R1; and then
 - (iii) separate the DNA fragments on an agarose gel.

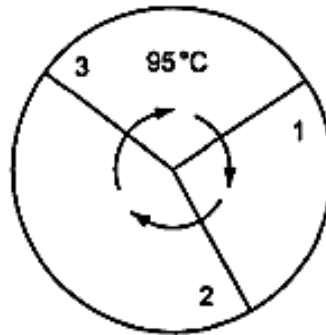
The results of such an experiment are shown in the figure below, using DNA obtained from each of the five family members.



Which family members have the disease?

- A A and B
- B C and E
- C D only
- D A, B and D

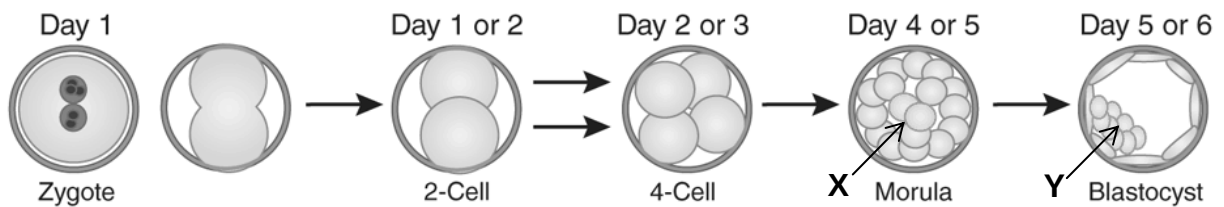
35 The diagram represents the stages of a polymerase chain reaction.



Which row shows the events happening at stages 1, 2 and 3?

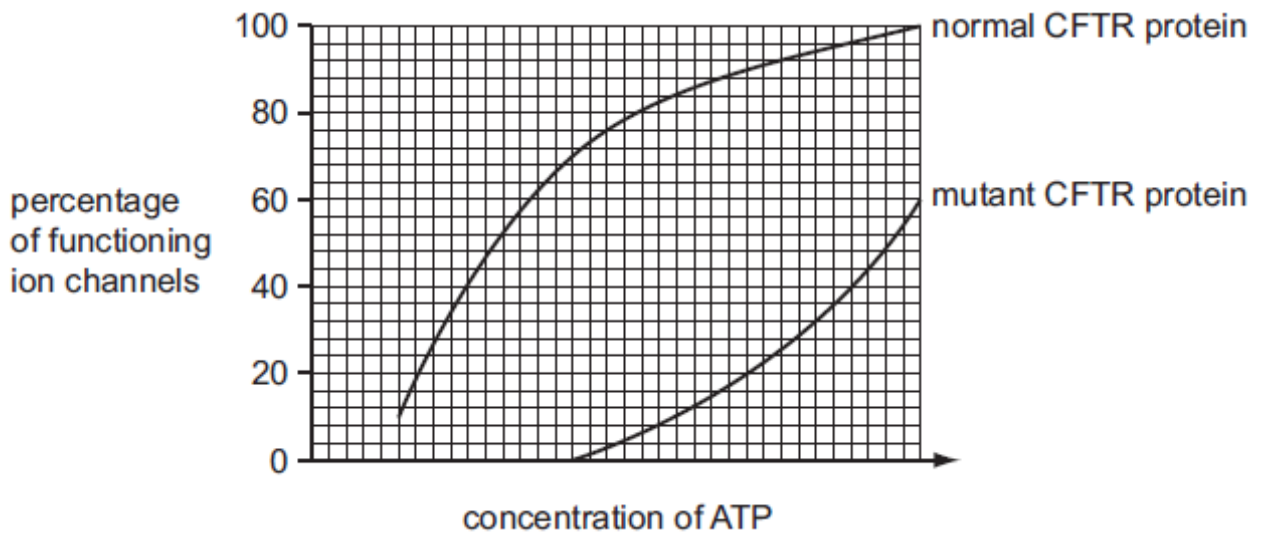
	1	2	3
A	binding of primers at 55°C	polymerization at 72 °C	denaturing of DNA
B	denaturing of DNA at 72°C	binding of primers at 55°C	polymerisation
C	polymerization at 55 °C	binding of primers at 72°C	denaturing of DNA
D	polymerization at 55 °C	denaturing of DNA at 72°C	binding of primers

36 Which of the following statements are true about the cells labelled X and Y?



- A X is a pluripotent cell while Y is a multipotent cell.
- B X is a totipotent cell while Y can give rise to multipotent cells.
- C Y will develop into the entire foetus including its placenta.
- D Y will give rise to pluripotent cells but X can only give rise to totipotent cells.

- 37 One of the many recessive mutations of the CFTR gene changes one amino acid in the region of the CFTR protein that binds ATP. The graph shows the effect of different concentrations of ATP on normal and mutant CFTR proteins.

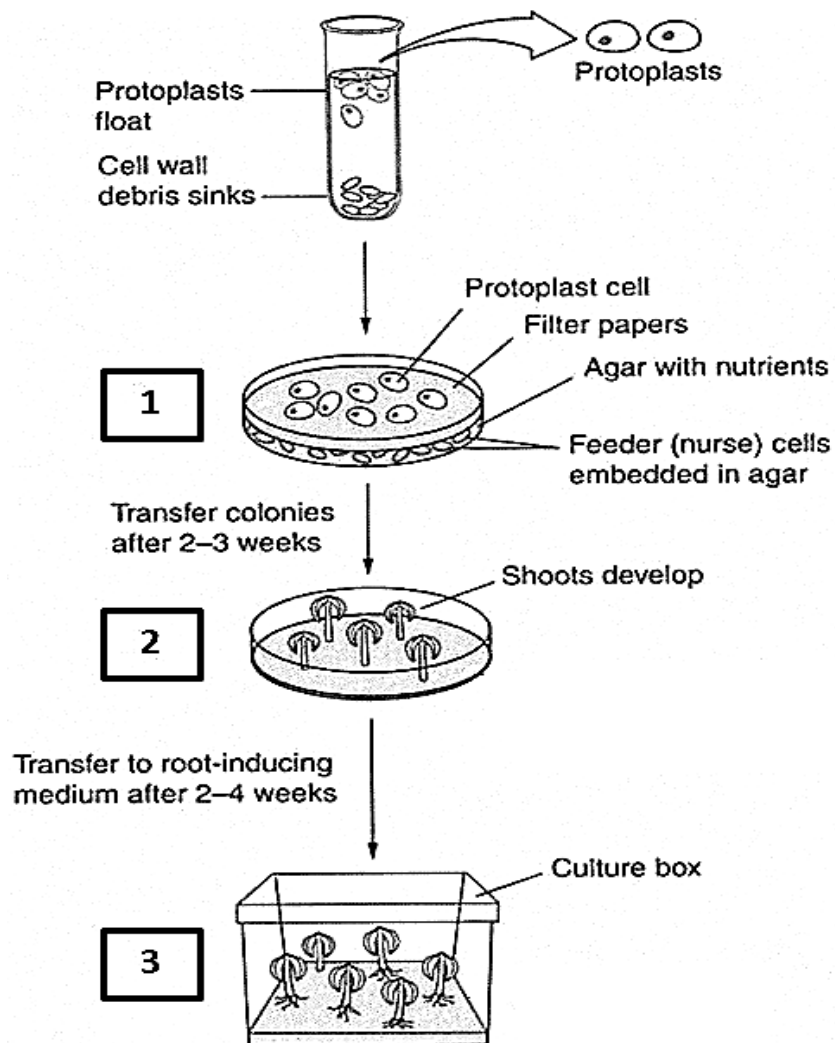


Which correctly describes individuals who are homozygous for this mutation?

- 1 Their CFTR protein cannot bind ATP and cannot act as an ion channel.
- 2 Their CFTR protein binds ATP less readily than normal CFTR protein.
- 3 They produce CFTR protein that must bind ATP to function as an ion channel.
- 4 They produce a mixture of normal and mutant CFTR protein, both of which can act as an ion channel.

- A** 1 only
- B** 2 only
- C** 2 and 3 only
- D** 2 and 4 only

- 38 The diagram below shows steps of the procedure that can be used to produce plantlets from protoplast cells isolated from leaves.



Which of the following auxin : cytokinin ratios are correctly employed in growth medium for the respective stages of plant tissue culture?

	1	2	3
A	2 : 0.5	2 : 0.2	2 : 0.02
B	2 : 0.2	2 : 0.02	2 : 0.5
C	2 : 0.02	2 : 0.2	2 : 0.5
D	2 : 0.2	2 : 0.5	2 : 0.02

- 39** Developing fish eggs can be treated to produce a diploid egg. In salmon, such eggs have been fused with haploid salmon sperm to give infertile triploid salmon.

Reproductive organ tissue from diploid trout was transplanted into newly hatched triploid salmon. This tissue matured as the fish grew and the salmon successfully produced viable trout sperm or eggs which resulted in young trout.

Which fish(es) could be seen as genetically modified?

- 1 Salmon providing eggs for treatment
- 2 Trout providing reproductive organ
- 3 Young triploid salmon
- 4 Young trout

- A** 3 only
- B** 1 and 3 only
- C** 3 and 4 only
- D** All of the above.

- 40** Which can best be considered to be a social implication of genetically modified (GM) crop plants?

- A** There is evidence that genes for herbicide resistance have spread from GM oilseed rape to nearby weed species although there is no evidence that these hybrids have persisted in the environment.
- B** In some studies, the number of beneficial insects has been reduced in areas where insect resistant GM Bt corn has been grown, however in other studies, there has been no effect or the number of beneficial insects has increased.
- C** Pressure placed by people on a multinational GM seed company has caused them to pledge that they will never develop a 'suicide' gene that prevents farmers from growing seed collected from last year's crop.
- D** In some countries, government regulations have been put in place to prevent farmers planting GM crops close to non-GM and organic crops to prevent GM pollen from reaching the crops.

PAPER 1 ANSWERS

1	A	2	D	3	B	4	D	5	D
6	B	7	A	8	C	9	A	10	C
11	C	12	A	13	C	14	D	15	D
16	B	17	C	18	B	19	B	20	D
21	B	22	A	23	D	24	C	25	C
26	B	27	D	28	B	29	D	30	D
31	B	32	A	33	C	34	B	35	A
36	B	37	C	38	D	39	A	40	C